

Week 8: Learning Theories

Design Thinking, Indigenous Pedagogies, and Revolutionary Frameworks

Introduction

This week's materials pushed me to consider learning theories that move beyond Western psychological traditions—frameworks that center embodiment, Indigenous ways of knowing, and revolutionary rather than evolutionary change. Reading Ward et al.'s history of enactivism alongside Resnick's constructionism, then pivoting to Aikman and Rahman's work on Indigenous pedagogies, created a powerful tension: What does it mean to 'design' learning when design itself carries colonial assumptions about control, prediction, and expert-driven change?

The design frameworks slide deck added another layer, inviting me to consider where my own pedagogical commitments sit among approaches like Culturally Responsive Pedagogy, Critical Fabulations, and Rasquachismo. Throughout this week, I found myself asking: How do we honor diverse ways of knowing while working within systems that were never designed for liberation?

Ward, Silverman & Villalobos: The Varieties of Enactivism

Core Concepts

Ward, Silverman, and Villalobos trace enactivism's emergence as a revolutionary challenge to cognitivism's 'mind as computer' paradigm. The article opens with an important epistemological claim that resonates with critical pedagogy: 'considering how a relevant chunk of intellectual history appears from a particular perspective helps in understanding that perspective and its motivations' (p. 365). This meta-awareness—that all theory-building is perspectival and motivated—echoes Freire's insistence that there is no neutral education.

Enactivism rejects representationalism (the idea that cognition involves mental representations of an external world) in favor of seeing cognition as

embodied

(grounded in sensorimotor systems),

embedded

(situated in physical and social contexts), and

enacted

(co-created through organism-environment interaction). Knowledge isn't 'acquired' or 'transferred' but emerges through dynamic coupling with the world.

Critical Reflections

What struck me as both powerful and troubling is how enactivism's revolutionary rhetoric doesn't fully interrogate whose perspectives dominate its intellectual history—three male authors tracing a genealogy of predominantly male, Western theorists (Varela, Thompson, Brooks, Gibson, Heidegger, Merleau-Ponty). Where are the

voices of marginalized learners whose embodied, enacted ways of knowing have been systematically excluded from cognitive science?

The article felt circular and repetitive after a time, which I think reflects enactivism's tendency toward philosophical abstraction. While the emphasis on embodiment and context is crucial, I found myself wondering: What does enactivism look like in practice? How do we create learning environments that honor embodied knowing without simply romanticizing 'hands-on' activities or appropriating Indigenous pedagogies that enact these principles far more fully than Western education ever has?

Connections to My Practice

Enactivism provides theoretical grounding for why technology access matters so profoundly: when students lack reliable internet or devices, they're denied participation in extended cognitive systems that constitute contemporary learning. My tech-based autoethnography project must attend to how educators' technological literacies are embodied, embedded in institutional contexts, and enacted through actual practice—not just abstract 'beliefs' about technology.

Resnick: Constructionism and the Four (Five) P's

Core Concepts

Mitchel Resnick's interview discusses his work extending Seymour Papert's constructionism through Scratch and the Lifelong Kindergarten group at MIT. Resnick articulates the Four P's framework for creative learning: Projects, Peers, Passion, and Play. He emphasizes 'computational fluency' over 'computational thinking'—shifting from problem-solving abstraction to empowering learners to express themselves, share ideas, and build identity through technology. A key distinction Resnick makes is between 'learning to code' (instrumental skill acquisition) and 'coding to learn' (using programming as a medium for thinking, creating, and expressing). Scratch's emphasis on remixing, collaborative creation, and project diversity reflects this philosophy.

My Revolutionary Stance

I noted in the article that I am firmly camp **revolution**, not evolution. Resnick acknowledges that 'many of those seeds sprouted and were stomped on by systems that made it difficult for them to keep growing' (p. 363). This is crucial: technology alone won't transform education. We must actively dismantle the systems that domesticate revolutionary tools for compliance and control.

I'm concerned about the 'Trojan Cat' strategy Resnick describes—using programming as a 'stealth' way to teach math and literacy. While I understand the pragmatic need to work within existing systems, this risks reinforcing that technology's value lies in its instrumental use for traditional academic outcomes, not in its potential to fundamentally reshape what counts as learning and knowing.

Critical Questions

Who has access to 'kindergarten' learning throughout life? Resnick's vision assumes resources, time, and institutional support for exploratory, passion-driven work. Yet

the very students who would benefit most from this approach—those from marginalized communities—face schools obsessed with test prep, behavior management, and 'catching up' to dominant norms. How do we design for liberation when the system's purpose is domestication?

Additionally, I worry about Scratch's evolution from 'expressive medium' to 'coding curriculum.' When programming gets stripped of creativity and community to fit into existing school structures, we've lost what made it revolutionary. The Four P's require systemic change, not just better tools.

Aikman: The Arakbut and Education Under Global Capitalism

Core Concepts

Sheila Aikman's ethnographic study of the Arakbut people in Peru's Amazon region documents how global capitalism—particularly gold mining—destroys both the physical landscape that makes Indigenous knowledge possible and the social fabric that transmits it across generations. This isn't just environmental destruction; it's epistemological violence.

Aikman describes traditional Arakbut education as fully integrated with daily life: children learn through observation, participation, and storytelling embedded in seasonal rhythms, kinship relations, and spiritual practices. Knowledge isn't abstract or decontextualized—it's *always* relational, tied to place, and enacted through practice. The article traces how formal schooling—even when well-intentioned—fractures these lifeways by removing children from learning contexts, privileging literacy over oral tradition, and positioning Western knowledge as superior.

Critical Reflections

This article was hard to read. Confronting the harm perpetuated on people in the name of capitalism and 'development' is devastating. But it's also essential. Aikman forces us to see that our education system has done similar violence—just more slowly and with cleaner hands. We've destroyed the social and relational landscape that makes meaningful learning possible, replacing it with isolated individuals competing for credentials in service of economic productivity.

What resonates deeply is Aikman's refusal to romanticize pre-contact Arakbut life while still insisting their educational practices hold profound wisdom. This isn't about returning to a mythical past—it's about asking what we've lost and what we might recover. The Arakbut model shows learning as integrated, communal, and grounded in reciprocal relationships with land and community. Western schooling fragments this into isolated subjects, individual achievement, and abstract knowledge.

Connections to My Own Experience

Reading this, I thought about my own childhood learning fractions. I grew up learning to crochet and knit with my grandmother, learning to woodwork with my grandfather, baking and scaling recipes with my mother. When we studied fractions in school months later, it made perfect sense—not because I memorized procedures, but because fractions were already integrated into my lived experience. What if all learning looked like that? What if we trusted students' lived knowledge as the foundation rather than treating them as empty vessels?

Rahman & Cochrane: Indigenous Pedagogies and Informal Learning

Core Concepts

Rahman and Cochrane's editorial introduction explores how Indigenous pedagogies challenge dominant Western educational frameworks. They emphasize the tension between formal schooling's emphasis on standardization, assessment, and predetermined outcomes versus Indigenous approaches that value *emergent learning, relational knowing, and attention to context*.

Key themes include:

- **Learning that 'overshoots its objectives'** — allowing for emergent, unpredictable growth beyond predetermined goals
- **Attention-based learning** — emphasizing deep observation, patience, and relational engagement rather than information transmission
- **Peripheral participation** — legitimate learning happening at the edges through observation and gradual involvement
- **Relational epistemology** — knowledge as always situated in relationships with community, land, and ancestors
- **Education for the 'good life'** — learning oriented toward flourishing rather than economic preparation

Critical Reflections

I highlighted the line about learning that 'overshoots its objectives' because it captures everything wrong with our obsession with learning outcomes, measurable objectives, and predetermined competencies. What if the whole point is that we *can't* predict or control what learning will emerge? What if trust in emergence is pedagogically necessary?

But I also noted: 'Yet so hard to let go of total control.' This is my struggle as an educator working within systems that demand evidence of learning, alignment with standards, and measurable progress. How do you create space for attention-based, emergent learning when the institution requires you to justify every moment with a learning objective? This tension feels impossible to resolve without systemic change.

Connections to My Practice

Rahman and Cochrane's emphasis on 'ethnographic sensibility' and learning from mistakes resonates with my autoethnography project. What if my website isn't about transmitting 'best practices' but creating space for educators to attend to their own experiences, develop critical consciousness about technology, and engage in relational dialogue about their practice? That would honor Indigenous pedagogical principles while working within (and against) Western educational structures.

Design Theories Slide Deck: Mapping My Pedagogical Profile

Framework Overview

The slide deck presented various learning design frameworks and invited us to consider where our own pedagogical commitments align. Frameworks included Culturally Responsive Pedagogy, Critical Fabulations, Rasquachismo, Universal Design for Learning, Critical Race Theory, and others. Rather than positioning these as neutral tools to 'apply,' the deck emphasized that design frameworks embody particular values, power relations, and assumptions about learning.

My Profile

I created a heatmap visualizing where my pedagogy aligns across six dimensions:

- **Liberatory/Critical Focus** — High alignment with frameworks that explicitly challenge oppressive systems
- **Emergent/Flexible Design** — Strong resonance with approaches that trust emergence over predetermined outcomes
- **Margin-Centered** — Deep commitment to centering marginalized voices and ways of knowing
- **Relational/Collective** — Emphasis on community, dialogue, and collective meaning-making
- **Justice-Oriented Action** — Learning as preparation for transformative social action
- **Epistemological Pluralism** — Honoring diverse ways of knowing beyond Western rationalism

Critical Reflections

Creating this profile made visible how my commitments cluster around frameworks that resist standardization, center marginalized perspectives, and position education as inherently political. I scored highest on Critical Fabulations, Indigenous Pedagogies, Rasquachismo, and Culturally Sustaining Pedagogy—all frameworks that explicitly challenge dominant power structures and honor non-dominant ways of knowing.

What this exercise revealed is that my discomfort with instrumental, technical design approaches isn't just personal preference—it reflects fundamentally different assumptions about what education is *for*. ADDIE, SAM, and similar models assume education serves to efficiently transmit predetermined knowledge. The frameworks I align with assume education should disrupt, liberate, and honor plural ways of knowing. These are incompatible goals.

Synthesis: Design, Decolonization, and Revolutionary Pedagogy

This week's readings converge on a crucial question:

Can we design for liberation, or is 'design' itself a colonizing practice?

The Limits of Design Thinking

Ward et al.'s enactivism, Resnick's constructionism, and the design frameworks deck all emerge from Western intellectual traditions that position the designer/educator as expert who creates optimal learning conditions. Even progressive frameworks like UDL or CRP can be co-opted into technical solutions that don't challenge

fundamental power relations. Rahman and Cochrane's Indigenous pedagogies and Aikman's ethnography reveal what's missing:

trust in emergence, attention to relationship, and willingness to let learning overshoot our objectives.

Indigenous approaches don't 'design' learning in the Western sense—they create conditions for participation in ongoing life practices. Knowledge emerges through relationship, not through expert-designed interventions. This fundamentally challenges the entire enterprise of instructional design.

Embodiment and Technology

Enactivism's emphasis on embodied, embedded, enacted cognition helps explain why Aikman's account is so devastating. When gold mining destroys the forest, it doesn't just eliminate resources—it destroys the material-social-spiritual context that makes Arakbut knowing possible. Similarly, when schools remove children from community life, they're not just changing location—they're fracturing the embodied, relational contexts where learning happens.

Resnick's vision of computational fluency must attend to this: technology isn't neutral infrastructure for expressing ideas. Which technologies we have access to, how they're designed, and whose purposes they serve shapes what's thinkable and knowable. My autoethnography project must examine how educators' technological literacies are embodied—shaped by institutions, available tools, physical spaces, and power relations.

Revolution vs. Evolution

I keep returning to my annotation: 'I am firmly camp revolution.' Reform within existing systems—whether through better design frameworks, progressive pedagogies, or technology integration—preserves the fundamental assumption that education serves to prepare individuals for economic participation. Indigenous pedagogies and critical frameworks challenge this entirely: education should support communities to flourish, resist domination, and define the good life on their own terms.

But working for revolution while employed by institutions designed for domestication creates impossible contradictions. How do I honor emergent learning while meeting institutional demands for measurable outcomes? How do I center marginalized ways of knowing while teaching within disciplines that position Western knowledge as universal? There are no easy answers—only the commitment to keep asking these questions and building alternative structures where possible.

Toward a Liberatory Design Practice

If I'm going to engage in 'design' at all, it must be design that:

- Refuses predetermined outcomes and trusts emergence
- Centers marginalized knowledges and actively works to dismantle epistemic injustice
- Attends to embodiment, relationship, and context rather than abstract principles
- Builds capacity for collective action and systemic transformation
- Recognizes technology as political and attends to whose purposes it serves

This week's readings make clear that this kind of design practice requires ongoing critique, humility about the limits of Western frameworks, and commitment to

learning from traditions—especially Indigenous traditions—that have sustained liberatory pedagogy for millennia.

Questions for Reflection and Discussion

Discussion Question for Class

Rahman and Cochrane describe learning that 'overshoots its objectives'—emergent, unpredictable growth beyond predetermined goals. Yet our institutions demand learning outcomes, measurable objectives, and evidence of student progress. Is this tension resolvable through better design practices, or does it require fundamental transformation of how we structure education? What might it look like to build educational systems that genuinely trust emergence while maintaining accountability to learners and communities?

Personal Reflection Questions

Reflection 1: My Embodied Learning

When have I experienced learning as embodied, relational, and embedded in practice—similar to Aikman's description of Arakbut education or Rahman's attention-based learning? What conditions made that possible? Conversely, when has my learning felt most abstract, disconnected, and decontextualized? What does this reveal about the structures and pedagogies that support or inhibit meaningful learning?

Reflection 2: My Design Commitments

Looking at my design framework profile, I see clear alignment with liberatory, margin-centered, emergent approaches. Yet I work within institutions that value efficiency, standardization, and measurable outcomes. How do I navigate this tension in my daily practice? When have I compromised my values for institutional compliance, and when have I resisted? What would it take to create more space for revolutionary rather than evolutionary change?

Final Project Questions & Idea Stems

Question 1: Designing for Emergence

How might I design my tech-based autoethnography website to embody Indigenous pedagogical principles—particularly attention-based learning, relational knowing, and trust in emergence? Rather than presenting a linear narrative or prescriptive 'best practices,' could the site invite educators to slow down, attend to their own embodied experience with technology, and engage in dialogue that generates unpredictable insights? What features, structures, or design choices would support learning that 'overshoots its objectives'?

Question 2: Technology as Embodied Practice

Enactivism emphasizes that cognition is embodied, embedded, and enacted through organism-environment coupling. How might I frame technological literacy not as abstract skills or 'digital competencies' but as embodied practices shaped by institutional contexts, available tools, physical spaces, and power relations? What research methods would help surface how educators' relationships with technology are lived, felt, and enacted in specific material conditions? How do I avoid extracting 'best practices' that erase context?

Question 3: Whose Knowledge Counts?

Aikman's work reveals how formal education erases Indigenous knowledge systems while positioning Western frameworks as universal. How do I ensure my project doesn't reproduce this violence by centering academic/Western frameworks for understanding technology integration? How can I honor the diverse ways educators—particularly those from marginalized communities—already engage with technology, even when those practices don't align with dominant narratives about 'effective' technology use? What would it mean to position educators as experts on their own technological practice?

Question 4: Revolution or Reform?

Resnick's 'Trojan Cat' strategy uses programming to 'sneak in' deeper learning within existing school structures. But does this risk co-opting revolutionary tools for institutional purposes? How do I position my project—as reform that helps educators 'integrate technology better' within current systems, or as revolutionary practice that builds capacity for educators to critique and resist how technology reinforces surveillance, standardization, and control? Can a doctoral project do both? Should it try?

Idea Stem: Decolonizing Digital Literacy

What if 'digital literacy' isn't about mastering tools but about developing critical consciousness regarding how technology embodies particular values and power relations? My project could explore how educators interrogate the technologies they use—asking whose purposes they serve, what assumptions they embed, which ways of knowing they privilege, and how they might be resisted or redesigned. This shifts from technology integration to technology critique and transformation.